



Good mathematics education at the school level, and an emphasis on computer science concepts rather than solely on programming languages, are necessary to meet the demands of a trained, quality workforce"

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On the other hand, client-side programming consists of scripts that are designed to enable user interaction in the Web browser. Client-side programming includes verifying user data, making graphics dynamic, and making navigation easy. Actions performed within the browser, such as the pressing of buttons or filling forms, are responded to by sending a request for information to the Web server and displaying the received data. Client-side programming is responsible for all such behaviour of the browser.

"Different markup and scripting languages are used for server-side and client-side programming. One has to choose which side one wants to master, right from the beginning," says Amit Shah, software engineer with Directi Group, which offers mass-market Web products and services. Shah's advice is the starting point for anyone looking to Web programming as a career option.

HTML—HyperText Markup Language—forms the base platform to define the layout and structure of a Web document. HTML is complemented by various languages—PHP (Hypertext Preprocessor), ASP (Active Server Pages), XML (eXtensible Markup Language), and more.

With PHP, for example, it is possible to create levels of security against malicious attacks. Using ASP can make Web pages dynamically editable.

Languages

Scripting languages are relatively simple programming languages that are used to create scripts that, when inserted into a Web page, control various elements of the page, such as the user interface, styles, and HTML markup. The scripting language interprets requests for HTML documents and objects embedded in the HTML.

PHP and ASP are some of the widely-used scripting languages. They can result in a code base as much as ten times smaller for the solution for a problem (as compared to a "regular" programming language), which increases the programmer's productivity by reducing the number of bugs in the code.

Scripting languages are of two types—client-side and server-side. Client-side scripting refers to the use of scripting languages such as JavaScript to automate applications that are executed on the client side (within the user's browser). Server-side scripting involves executing scripts on the Web server before any data is passed on to the browser.

Server-side Or Client-side?

A person good at networking and with a ready hand on hardware and Web protocols is best suited for server-side programming. Individuals with

good knowledge of databases and servers can opt to learn languages like Perl and Python for writing code for the Web server to perform object-oriented functions such as e-mail processing, advanced content management systems, general-purpose templates, and more.

Perl (Practical Extraction and Reporting Language) is widely used for writing programs for Web servers involving tasks such as generating reports, database synchronisation, auto-updating of user accounts, etc. Python is used for writing basic CGI (Common Gateway Interface) scripts, advanced content management systems, Web application frameworks, processing e-mail and RSS feeds, etc.

Those who wish to be in client-side programming need excellent knowledge of browser technology, operating systems, Internet protocols and such, with the ability to format content received from the Web server in the best possible manner.

The third option—that of working with databases—can be taken up by those from a database background. For more on careers in database, refer to *Digit*, April 2007.

The Tools Of The Trade

An education with a specialisation in computer science increases the chances of selection for a Web programming job. Students with an engineering, IT, or mathematics background are considered the best fit for Web programming by the industry. Graduates and post-graduates having majored in computer science (BE / ME in computer science), or those with a degree (BSc / MSc) in mathematics or statistics enjoy preference over candidates from other backgrounds.

"The industry prefers BE candidates with experience in development languages, and those with a diploma from reputed computer institutes with a good academic background. Also, candidates from a science background with computing or mathematics as a major subject are also given preference," says Harish Sukhwil, CEO, Hashtech Systems Pvt Ltd, which offers Web services including hosting, business process outsourcing, and security solutions.

Sukhwil also says that the skill set of a good programmer in today's context should ideally be one of the following:

- ☐ PHP + MySQL + XML
- ☐ ASP + MSSQL + XML
- ☐ ASP.NET + MSSQL + XML + AJAX
- ☐ JSP + Oracle + JavaBeans

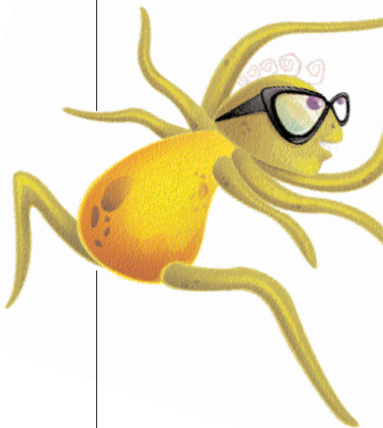
Candidates from other disciplines such as mechanical engineering, electrical engineering, chemical engineering, and others streams are also recruited, though.

"Information technology as a part of the curriculum for varied disciplines, such as commerce and arts, will ensure professionals are better prepared for the challenges of online business," says Venki Nishtala, CTO, Rediff.com.

The perfect destinations to get equipped with the tools of the trade are universities and colleges approved by the All India Council of Technical Education (AICTE).

There are also private institutes that offer a number of short-term or one-year courses, which can be pursued for an additional qualification. They provide education and training in specific products (such as Dreamweaver, Fireworks, and FrontPage) and the syntax of languages like HTML, PHP, ASP, and Java, to name a few.

The quality of education at a private institute, however, depends upon the teaching facul-



The Languages

Base platforms for Web page development	HTML, XML, PHP, ASP
Basic Programming	C, C++, PHP, Java, Perl, Python
Visual Programming	Visual C++, Visual Basic, Java, ASP.NET, Java Server Pages (JSP), AJAX (Asynchronous JavaScript and XML)
Database Systems	SQL (Structured Query Language), MySQL, Oracle, MSSQL (Microsoft Structured Query Language)